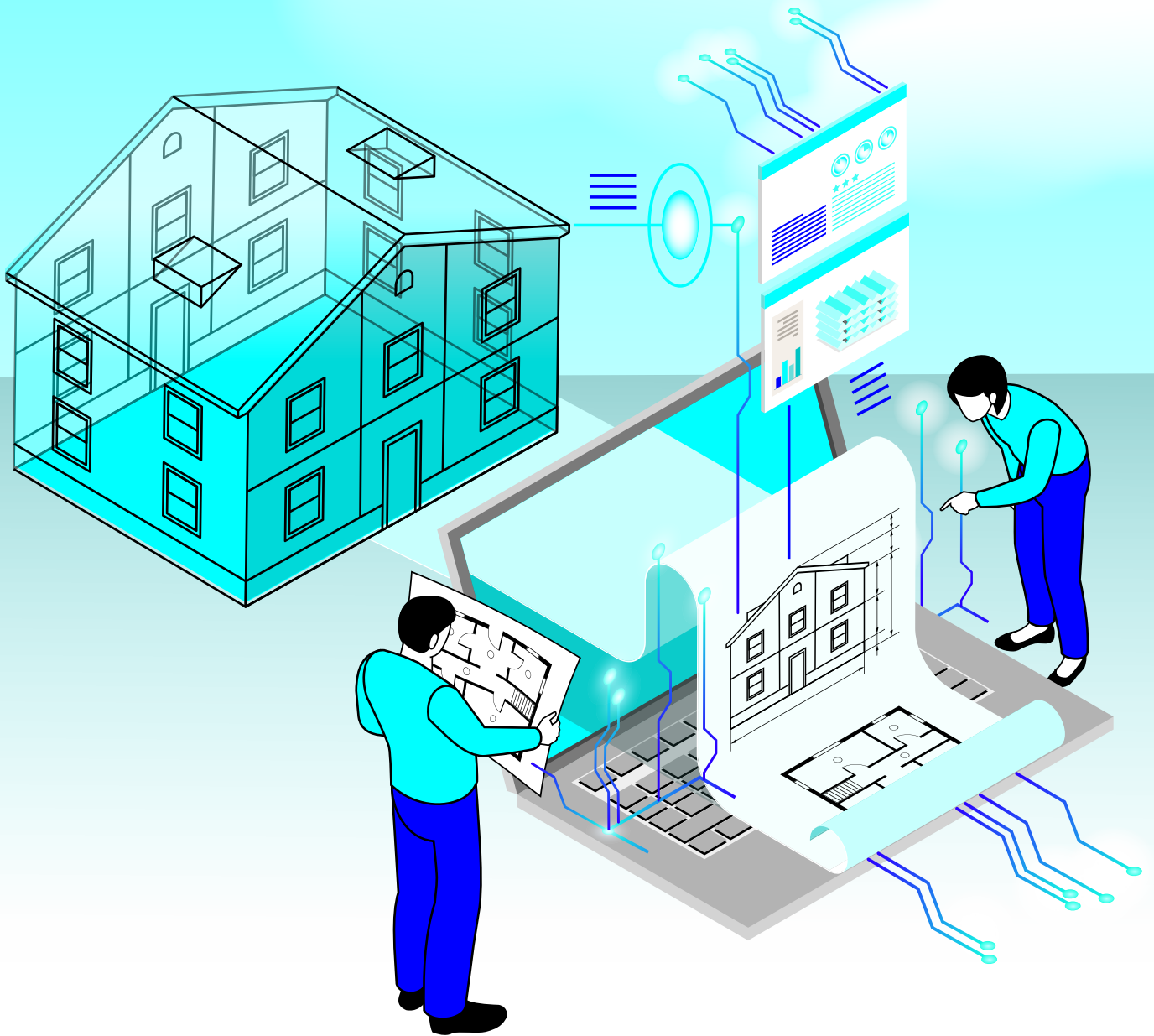




AI for building permits

*Use cases, prototypes
and practical insights*

The planning permission procedure is a complex administrative process. Complying with numerous building regulations is challenging, but incomplete or incorrect planning applications increase the workload even further. This poses significant challenges for municipalities, specialist agencies and applicants. This sandbox project investigated the use of artificial intelligence (AI) to optimise the building permit process and to provide targeted relief for those involved. Together with practical and technology partners, the Office for Economic Affairs of the Canton of Zurich tested specific AI use cases and conducted a systematic AI analysis. The analysis focused on evaluating the current capabilities of different AI models and revealed significant differences in the consistency and quality of the results, particularly in the interpretation of plan representations. The development of the prototype for AI in the reporting process (e.g. for solar panels, heat pumps or charging stations) showed that rule-based systems achieve high efficiency gains, while AI-based pre-checks provide added value. The report contributes to the future use of AI for building permits. It shows where AI can provide concrete support today, where rule-based solutions are preferable and where there are currently practical limitations. At the same time, it makes clear why technological innovation must always be developed in the context of legal, organisational and institutional frameworks.

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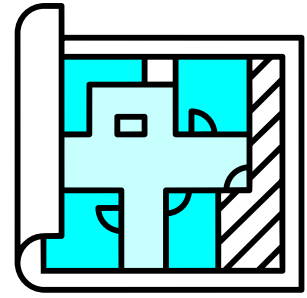
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01.

Building permit procedures are highly complex



The building permit procedure is a key tool for managing spatial development. It decides on building activity and use and ensures that public interests such as architectural and design quality, legal equality or the protection of neighbours and the environment are safeguarded. The high technical and legal complexity of the proceedings is an expression of this mandate. The process has been developed over decades and is based on numerous legal foundations as well as on established enforcement practices at cantonal and municipal level. The federal structure makes it possible to adapt to local conditions, but at the same time leads to a great deal of structural diversity.

Classification in the overall process

The building permit procedure is of great importance, but it is only one part of the construction process. In the overall duration of a construction project, it often takes a comparatively short period of time. Challenges such as delays can also arise in the planning, financing and execution of a construction project. The approval process should therefore not be viewed in isolation, but rather as part of a complex overall system with many actors and dependencies.

Challenges of the building permit process

The planning permission procedure is characterised by federal structures, high technical requirements and

varying organisational requirements. In the canton of Zurich, responsibilities are shared between the municipalities and the canton: The municipalities are primarily responsible for the building permit process, in selected areas the canton. Practical implementation shows recurring patterns that increase the workload, lengthen procedures and make it difficult to trace decisions:

- **Fragmented requirements**

The building regulations are fragmented. Numerous public and private interests are taken into account in the planning process, which are reflected in various legal bases. Cantonal guidelines meet municipal regulations and local practice (e.g. green space figures or parking calculations). The requirements vary depending on the building zone, building type and type of project. The processes and procedures of the respective authorities are only comparable to a limited extent, which inhibits automation and makes scaling more difficult.

- **Limited resources in administration**

The organisational requirements vary greatly. While larger cities with dedicated systems and specialist departments handle large numbers

01. Building permit procedures are highly complex

of cases, smaller municipalities often rely on external support and work with scarce human resources. Specialist knowledge and decision-making powers are often concentrated on just a few people, which can lead to uncertainties when it comes to application in the event of staff changes or increasing legal complexity.

- **Strict technical and legal requirements**

The physical examination of building applications requires in-depth legal expertise and experience, with an increasing shortage of skilled workers. Many standards are complex and require interpretation. Decisions are often based on empirical knowledge, which is poorly documented and difficult to transfer. The coordination of numerous specialist units (at federal, cantonal and municipal level) also increases the workload.

- **Incomplete and incorrect entries**

A significant proportion of planning applications are incomplete or incorrect. Formal deficiencies and missing information prevent a substantive review and lead to the suspension of proceedings, even though the planning intent is often clear. The quality of the documents varies greatly. Entries range from hand-drawn plans to complex digital models.

These challenges are interlinked. Local authorities are under increasing pressure: The average length of proceedings has increased in recent years,¹ and the proceedings tie up considerable human resources. At the same time, current developments in AI and digital administration are opening up new opportunities to

support processes in a targeted manner and to mitigate some of these challenges.

Ongoing digitalisation

Many cantons and municipalities have been relying on the gradual digitalisation of their building permit procedures for several years now. Digital entry portals, electronic dossiers and the first automation systems are well established in many places. In the canton of Zurich, for example, from 1 April 2027, all planning applications must be submitted via the «eBaugesucheZH» platform.² However, the digital transformation does not only affect public authorities. This change is also taking place in the planning and construction industry. The quality, structuring and standardisation of building applications depend largely on the planning tools and working methods used. Digital models (such as *Building Information Modelling [BIM]*), structured data and consistent documents are key prerequisites for more efficient processes.

Further development of existing processes

The diversity and complexity of the procedures reflects both federal structures and good practice. Standardisation and automation must therefore be targeted within existing structures where they create added value. At the same time, systemic issues need to be clarified at a political level in order to create reliable and coordinated framework conditions for further development (see also postulate «AI in the building permit procedure»³).

* Highlighted words are explained in more detail in the glossary.

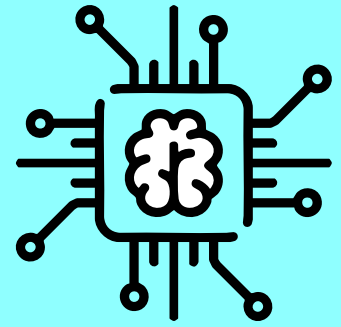
¹ Decision of the Council of Ministersaon postulate KR no. 226/2023

² Electro building applications | Canton of Zurich

³ Cantonal Council of Zurichch | Businesses | AI in building permit procedure

02.

Overview of AI use cases



Technological advances in the field of AI are opening up new opportunities for specifically addressing key challenges in building permit procedures. AI can help make processes more efficient, consistent and accessible – for both applicants and authorities. The aim is not to fully automate decisions, but to provide support throughout the entire process. Suitable systems are often based on the same AI models, in particular *large language models (LLMs)*. Depending on the application, they can be used differently, for example for text-based queries on building regulations or for interpreting plans. Various categories of use cases can be distinguished along the planning permission process.

1. Information and orientation

AI can make it easier to access information in building permit procedures in a targeted manner. *Semantic search methods* and *chatbots* make it possible to provide contextual information on procedures, requirements and building regulations. One approach is *Retrieval-Augmented Generation (RAG)*. The relevant systems access verified legal bases, information sheets and procedural documents and present relevant content in a structured and comprehensible manner. Currently, most of this information is distributed through numerous websites and isolated documents. AI systems automatically answer recurring questions and explain

requirements on a case-by-case basis. The aim is for applicants to receive orientation at an early stage and for municipalities to receive less repetitive or avoidable queries. Similarly, municipalities and specialist departments can use AI to better tap into and link internal information and make it available in a context-specific manner.

2. Preliminary review of the building application

Another use case begins even before the official submission of a building application. AI and rule-based automation enable structured preliminary verification of input documents. Digital portals record the plans and forms. Before submission, an AI system checks the documents for completeness, formal requirements and compliance with legal requirements, and provides applicants with targeted and comprehensible information on any need for improvement.

«The greatest leverage of AI currently lies in improving the quality and completeness of building applications before submission.»

*Raphael von Thiessen, AI Sandbox
Programme Manager, Canton of Zurich*

02. Overview of AI use cases

Thanks to early feedback, applicants can optimise their documents before submitting the building application. The aim is to improve the quality and completeness of applications even before they enter the formal approval process.

3. Support with official enforcement

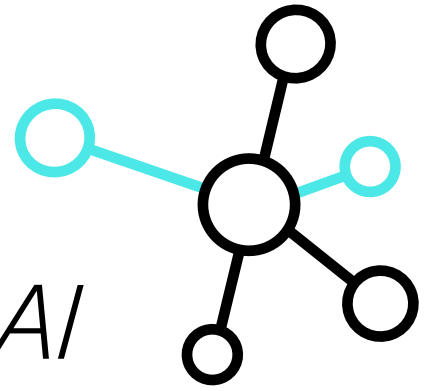
In the future, AI systems could also make a targeted contribution to reducing the burden on authorities by systematically checking formal criteria (e.g. completeness) and supporting substantive checks of building regulations. If the legal situation is clear, they could also take on preparatory sub-steps, such as the pre-formulation of decrees, the structured preparation and summary of facts, and the prioritisation of cases according to formal criteria. Decision-making authority, responsibility and traceability remain with the competent authority in all cases (see Chapter V).

Requirements for the use of AI

The effective use of AI in the building permit process requires appropriate technical, organisational and legal framework conditions. This includes adequate data quality, clear processes and responsible governance with human control. At the same time, the use of AI has clear limits: Not all questions can be solved technologically. In particular, legal fragmentation, fragmented system and application landscapes, discretionary decisions and participatory processes continue to require political-strategic clarification and a humane assessment. AI can provide targeted support for the building permit process, but it is no substitute for technical responsibility or political processes. The added value of AI can only be realised if it is embedded in a broader digital and institutional development process.

03.

Analysis of the potential and limits of AI



As part of the innovation sandbox for AI, the Office for Economic Affairs of the Canton of Zurich worked with project partners **NOKEMA** (project submission) and **Byte Studio** (technical development partner) to analyse the use of AI in the building permit process. Due to the high technical complexity of the application, the project team also worked with specialist partner **Gossweiler Ingenieure AG**. This partnership was crucial, as the ongoing exchange between the technological AI perspective and planning and approval practice and its specialist realities was a key prerequisite for achieving realistic results. In addition, with the consent of the applicants, the **city of Kloten** provided several real building applications and **registration procedures**. These served as a basis for systematically and practically examining the potential and limits of current AI models. However, since this is a single case study, the results can only be generalised to a limited extent.

Legal basis for AI analysis

The selection of the first use cases is based on clearly defined legal principles in the canton of Zurich. Relevant legal bases include in particular:

- Planning and Building Act (PBG), Canton of Zurich
- General Building Ordinance (ABV), Canton of Zurich
- Construction Procedures Ordinance (BVV), Canton of Zurich

These decrees form the legal framework for formal and substantive checks in the building permit procedure and serve as a reference for the issues investigated in the project.

Different stages of testing

Based on the aforementioned legal bases, the project focused on clearly defined, practice-relevant issues throughout the building permit process. These cover different test levels:

- **Procedural review (§ 14 BVV):** How will the construction project be handled?
- **Completeness check (§ 3 BVV):** Is the building application complete?
- **Formal review (§ 4 BVV):** Does the presentation of the plan comply with legal requirements?
- **Physical examination (various BVV sections):** How should the contents of the plan and the other application documents be evaluated in light of the relevant building law requirements?

These questions represent typical test steps in enforcement and are suitable for systematically investigating the potential of today's AI models.

AI analysis process

The audit was carried out in a clearly structured, multi-stage process. Based on the real entries for the construction of an apartment building, the project team defined specific test questions and defined a correct reference answer for each question. This served as the basis for the subsequent evaluation of the AI results. The team then technically prepared the construction plans. Depending on the AI

03. Analysis of the potential and limits of AI

model, the plans were either processed directly as PDFs or converted in advance into high-resolution image formats. In a next step, the team surveyed several AI models from different providers⁴ – including OpenAI, Google and Anthropic – with identical questions and identical material. This made it possible to directly compare the responses of the different models. The responses were evaluated using a separate AI model (*LLM-as-a-Judge*). This compared the responses generated with the reference solutions defined by the project team and automatically evaluated them. The tests repeated each query three times to find out whether a model consistently gave the same answer or whether the results varied randomly. Finally, the project team systematically evaluated the results and compared the tested models in terms of performance, response time and costs.

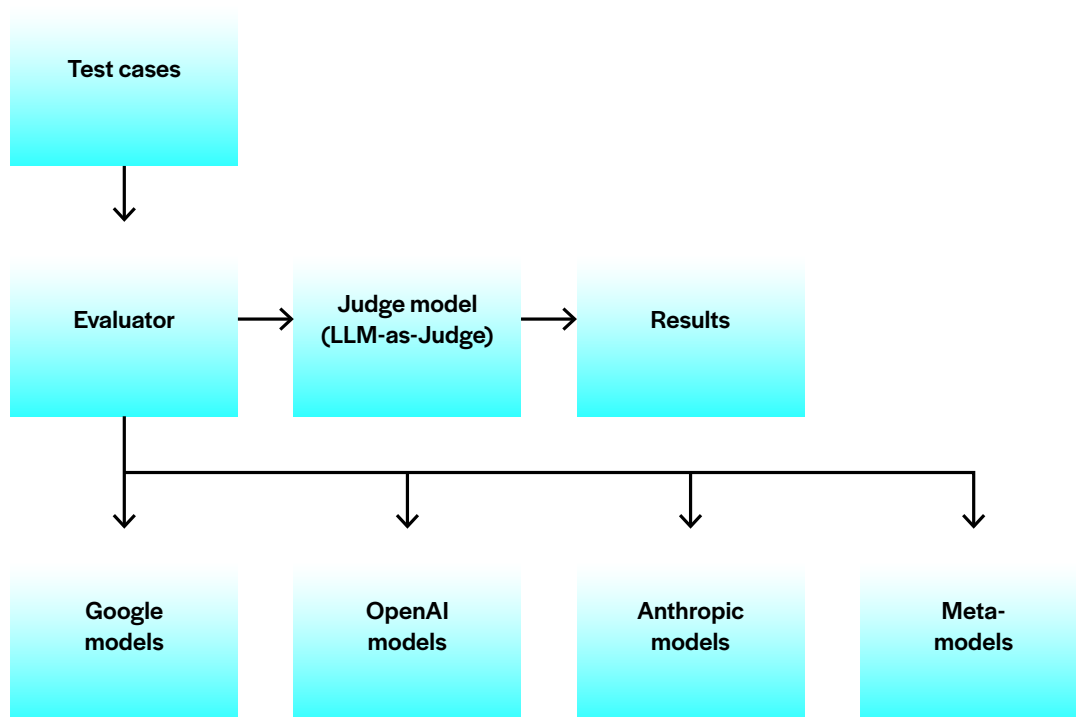


Figure 1: Conceptual structure of the tests with a benchmarking of different model providers

⁴ The following eight models were used for the AI analysis (grouped by provider): gpt-5-mini; gpt-5.1; o3; gemini-3-pro-preview; gemini-2.5-flash; gemini-2.5-flash-lite; claude-opus-4-5; claude-sonnet-4-5.

03. Analysis of the potential and limits of AI

Test design with five dimensions

A fully automated answer to the above questions in the form of a one-off model survey with correct output (zero-shot approach) is not realistic with current generative AI models. The legal and technical complexity of the issues, the multitude of relevant standards and inconsistent and in some cases unstructured input data lead to a high degree of contextual noise (noise). This makes it difficult to make a consistent and reliable assessment. The project team therefore divided the testing process into individual subtasks and tested these separately. The aim was to systematically analyse the capabilities and limitations of the models used along five clearly defined dimensions.⁵

1. Information extraction

Can the AI model specifically extract information from a plan, such as scale, number, date, architect, floor areas or building height?

2. Counting

Can the AI model recognise elements on a plan and correctly count them, such as windows, doors, rooms, parking spaces or apartments?

3. Measuring and calculating

Can the AI model work with masses and perform simple calculations, such as verifying area specifications or checking ratios such as that of the window area to the floor area?

4. Spatial understanding

Can the AI model record the arrangement and orientation of building components, such as the location of the entrance, the orientation of a balcony or the relationship between adjacent rooms?

5. Context understanding

Can the AI model correctly classify the plan and differentiate, for example, whether it is a floor plan or a section, which floor is depicted, or whether there is a new build or a conversion?



Figure 2: Example of a plan with which the answers to the questions were tested along the five evaluation dimensions

Results of the AI analysis

The evaluation of the 3.336 tests carried out clearly shows that AI can support the building permit process in a targeted manner, but a fully automated overall check is not realistic.⁶ The added value lies in clearly defined, standardised sub-tasks.

⁵ Note on test design: The AI models are evaluated using a score of 0 to 10 points. The score indicates how well the respective model answer matches the expected reference answers. A value of 10 indicates a complete match, a value of 0 indicates a completely incorrect answer in terms of content.

⁶ If interested, the technical report with the detailed results of the AI analysis can be requested from the Sandbox team.

03. Analysis of the potential and limits of AI

What already works well today

- **Information extraction:** AI models can extract structured information from plans if the resolution is sufficient. This includes area specifications, roof shapes, scales and simple metadata. However, the reliable measurement of distances (e.g. road or border distances) does not currently work consistently, as reference points, scales and presentation methods are interpreted differently by the AI models.
- **Context understanding:** The AI models provided reliable results for most examples (planty detection or type of environment, differentiating between green space, buildings and road).

Where the results are currently unclear

- **Information extraction:** AI models can extract structured information from plans if the resolution is sufficient. This includes area specifications, roof shapes, scales and simple metadata. However, the reliable measurement of distances (e.g. road or border distances) does not currently

work consistently, as reference points, scales and presentation methods are interpreted differently by the AI models.

What is currently difficult or unreliable

- **Measuring and calculating:** Complex measurement and calculation tasks with numerous plotted masses currently do not provide stable results; plausibility checks (e.g. whether the sum of partial masses corresponds to the total mass) do not work reliably. Simple measurement and calculation tasks in clearly defined scenarios – such as reading out an explicitly specified distance – can be reliably solved.
- **Counting:** Demanding counting tasks without additional information (e.g. windows in complex facades) are prone to error. On the other hand, counting tasks with clear specifications or markings work well, such as determining the number of basement compartments or fire doors with defined colour labelling.

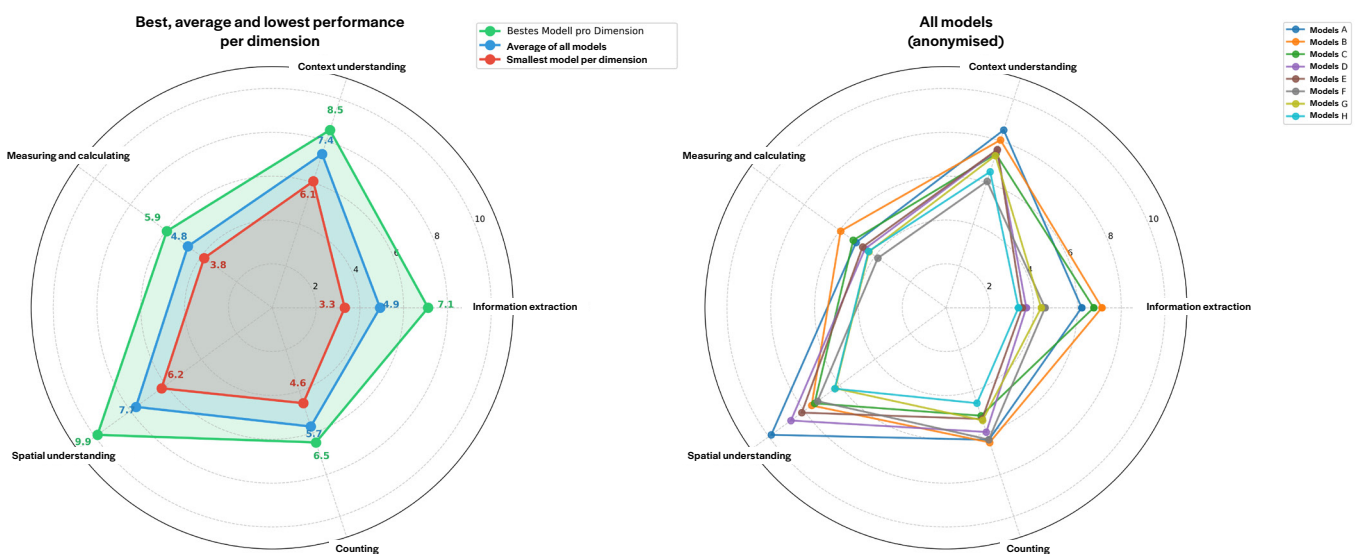


Figure 3: Performance of the AI models across the five dimensions. The best and the weakest model were evaluated separately for each metric (e.g. accuracy, speed). This means that the best model can be from a different provider depending on the metric (see diagram «All models»).

03. Analysis of the potential and limits of AI

Key findings for the use of AI

- **Using structured data:** Information that is already available in a structured form in digital planning models (e.g. BIM) should be checked there as far as possible on a rule-based basis. Subsequent evaluation of 2D PDFs is particularly useful when such data is not available and does not replace further digitisation of planning processes.
- **Input and resolution:** The quality of the analysis depends heavily on the input format (PDF or image), resolution and plan density; large or complex plans lead to loss of information due to technical resolution limitations.
- **Overall inspection and automation:** It is not possible to reliably verify a building application in a single step; AI is not suitable for end-to-end automation, but only for modular, step-by-step checks.
- **Division of tasks:** Many tests can be reliably implemented by breaking them down into clearly defined subtasks (e.g. identify legend → identify relevant elements → count).
- **Model differences:** Accuracy, consistency, speed and costs vary considerably between the different AI models and are constantly changing as technology advances. Statements on performance are therefore always time-dependent. Productive use requires continuous evaluation and the ability to exchange models flexibly.
- **Data management:** Consideration of the model provider and state control over administrative data is important. When AI models are used, personal data, land data, energy technology data and geodata from planning applications accessible online are further processed abroad and by private providers, depending on the model.
- **System architecture:** Regardless of the model choice, a modular, multi-level system architecture is required to handle context noise, resolution limits, and different plan formats, as well as to ensure scalability and maintainability.
- **Cost structure:** Costs per AI query range from around 0.1 to 2 centimes. An extensive preliminary check with many sub-steps and a model evaluation can currently be carried out for less than CHF 2 per building application.
- **Boundaries of complexity:** Most technical and legal assessments remain dependent on human expertise. For certain test steps that require very little time, the use of generative AI is in some cases questionable from a cost-benefit perspective, as implementation and monitoring can be high in addition to the costs per AI query.

«In AI-based processing of building plans, the choice between LLM-based and rule-based approaches, as well as their continuous evaluation, is critical.»

Yves Rutishauser, Founder and AI specialist, Byte Studio GmbH

03. Analysis of the potential and limits of AI

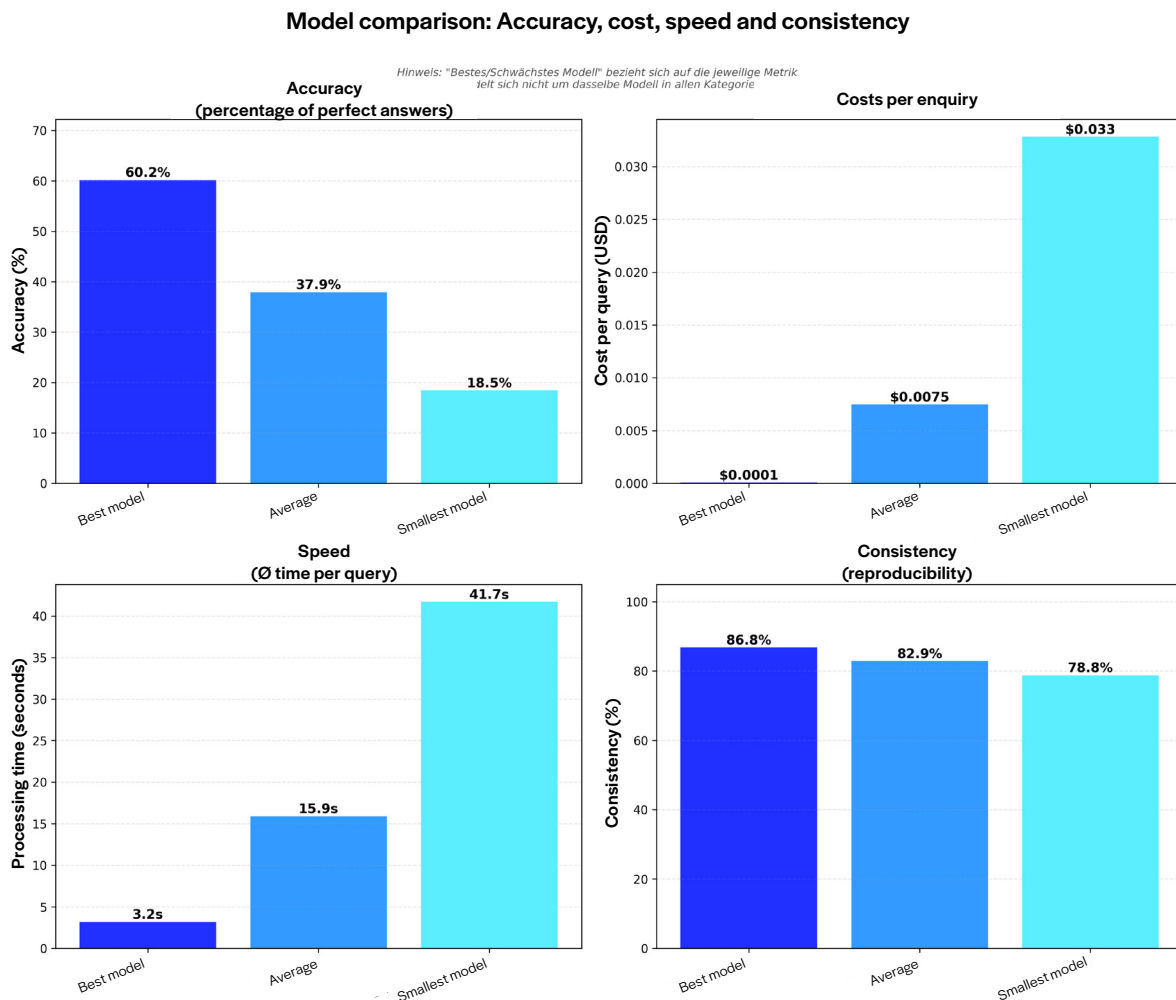


Figure 4: AucThe model comparison also revealed significant differences in the dimensions of accuracy, cost per query, speed and consistency: Depending on the dimension, different models achieved the best and worst results.

03. Analysis of the potential and limits of AI

The results show that AI can relieve the building permit process in a targeted manner, especially in formal and standardizable preliminary tests. The greatest benefit lies in AI-supported pre-checks, not in complete automation of the entire process. Prerequisites for productive use are clearly defined tasks, modular system architectures as well as ongoing quality assurance and evaluation of the models used.

Limitations of AI analysis

The present experiments are based on a limited number of building applications and therefore allow only limited generalisations. A robust assessment would require an extension of the evaluation to a larger data set with different project types (e.g. single-family homes, apartment buildings) and varying planning quality. For these reasons, the model comparisons in this report are presented anonymously. In addition, the listed capabilities of the systems relate primarily to objectively verifiable characteristics; evaluative assessments with discretion remain the responsibility of the responsible specialists and cannot be delegated.

Building on the findings and limitations, the project team developed a prototype in the next step to test the identified application potential under practical conditions.

04.

Prototype AI pre-check in the reporting procedure



The analysis of the AI potential has shown that the issues involved in the building permit procedure are technically and legally very complex. The project team therefore deliberately decided to narrow down the scope in order to develop a functioning and practical prototype. The focus was on the **notification procedure**.

—

Solar panels, heat pumps and charging stations

The starting point is a typical situation: Applicants are planning a construction project at a specific location that can generally be handled via the registration procedure, such as the construction of a solar installation, a heat pump or a geothermal probe. A total of eight different situations are foreseen, which are subject to the notification procedure. In practice, this raises two key problems.

Problem 1: Choosing the correct procedure

Choosing the correct procedure is time-consuming. An overview table specifies which procedure is to be applied based on the location-specific circumstances for the situations that are generally subject to reporting pursuant to § 2a BVV.⁷ It must also be checked whether the project is even permissible at the intended location (e.g. no geothermal exchanger heat pumps in the groundwater protection zones S1/S2).

⁷ Process description Registration Procedure Canton of Zurich

04. Prototype AI pre-check in the reporting procedure

Approval procedure (B) Municipal and cantonal responsibilities							B
Approval process (b) Municipal responsibility							b
Reporting procedure (M) Municipal and cantonal responsibilities							M
Reporting procedure (m) Municipal responsibility							m

Protective arrangements, inventories, spacing lines, usage zones	Solar panels on roofs (§2 a. Bst, a. BVV)	Solar panels on ^{façades} or free-standing (§2 a. b. + c. BVV)	Solar panels in industrial zones (roof, façade or free-standing) (§2 a. d. + e. BVV)	Indoor air/water Heat pumps (§2 a. f. BVV)	Outside air/water Heat pumps (§2 a. g. BVV)	Geothermal probe heat pumps (§2 a. h. BVV) ²	Connections to a district heating network (§2 a. in conjunction with BVV)	Publicly accessible charging stations (§2 a. j. BVV)
 supra-municipal urban landscape inventory	B	B	B	m	B	m ²	m	m
 supra-municipal conservation inventory, supra-municipal conservation order	B	B	B	m	B	m ²	m	m
 inter-municipal nature conservation objects and areas worthy of protection	M			m	M	m ²	m	m
 cantonal Landscape Protection Ordinance	M	M	M	m	M	m ²	m	m
 cantonal inventory of landscape protection properties	m	M	M	m	M	m ²	m	m
 municipal listed building inventory, municipal conservation order	b	b	b	m	b	m ²	m	m
 groundwater protection zone S1, S2	B	B	B	m	B			
 watercourses, bank strips	m	M	M	m	M	B	m	m
 construction line ⁴	m	M	M	m	M		m	m
 forest distance	m	M	M	m	M	B	m	m
 clearance zones	m	B		m	M	m ^{2,3}	m	M
 recreation areas	m	B		m	M	m ^{2,3}	m	M
 reserve zones	m	B		m	M	m ^{2,3}	m	M
 agricultural zones	m	B		m	M	m ^{2,3}	m	M
 hamlet area	m	B		m	M	m ²	m	M
 core zones	b	b		m	b	m ²	m	m
 quarter conservation zones	m	m		m	m	m ²	m	m
 central zones	m	m		m	m	m ²	m	m
 residential areas	m	m		m	m	m ²	m	m
 industrial and commercial zones	m	m	m	m	m	m ²	m	m
zones for public buildings	m	m		m	m	m ²	m	m

¹ For photovoltaic systems on façades, there is still no state-of-the-art paper on fire protection. Until that time (expected at the end of 2024), Swissolar's transitional document 'Fire protection for rear-ventilated photovoltaic systems on façades' will apply. The guidelines outline which façades can be used to plan and, if approved, implement photovoltaic systems for fire protection, under which conditions and with which corresponding verification method for fire protection.

Figure 5: Decision matrix according to the process description of the registration procedure in the canton of Zurich

04. Prototype AI pre-check in the reporting procedure

Depending on the type of project and site-specific conditions, four different procedures are used:

- **Approval procedure (B)**
Responsibility of the municipality and canton
- **Approval procedure (b)**
Responsibility of the municipality
- **Reporting procedure (M)**
Responsibility of the municipality and canton
- **Reporting procedure (m)**
Responsibility of the municipality

This assessment depends on a variety of protective arrangements, inventories, spacing lines and usage zones. These include supra-municipal landscape inventories, cantonal landscape protection ordinances, groundwater protection zones, forest and waterway distances and the respective usage zone. Several criteria may be relevant for a single location at the same time. In each case, the most time-consuming applicable procedure is decisive. The check is carried out on the basis of various thematic maps in the GIS browser as well as other legally adopted documents and is carried out on a case-by-case basis by a responsible specialist at municipal level.

Problem 2: incomplete and incorrect entries

The entries in the notification procedure are often incomplete or incorrect. Although the Construction Procedure Ordinance clearly stipulates which documents must be submitted, in practice documents are regularly missing or formal deficiencies occur, such as incorrect scales, missing drawings of relevant elements or incomplete planning details. This leads to queries, additional interactions and delays, even though the material scale of the projects is comparatively small.

Prototype AI pre-check in the reporting procedure

The prototype aims to provide support precisely in these early stages: in classifying the applicable procedure and in the formal preliminary examination of building applications. The following four steps summarise the main functionality of the prototype.

04. Prototype AI pre-check in the reporting procedure

Step 1: Capture project location

Users select the address of their project. The prototype shows the location in a map view and represents the immediate surroundings. This makes the spatial context of the project visible at an early stage. It forms the basis for the next steps.

 **Project location**
Enter address

 **Project details**
Select project type

 **Upload documents**
Add documents

 **Confirmation**
Review & conclusion

Location

Please search for the address of the project location.

Search address*

Selected address

Street	House number
Walchestrasse	19
Postcode	Town/city
8006	Zürich

Location on the map



The marked area shows the location of your project

Figure 6: Selection of an address for the project location (here, for example, the building of the Office for Economic Affairs, Walchestrasse 19, 8006 Zurich)

04. Prototype AI pre-check in the reporting procedure

Step 2: Select the project type and determine the process

Users select the type of their project from a predefined list of eight reportable circumstances. Based on this, the prototype determines the applicable procedure. The underlying process matrix depicts the legal requirements fully rule-based. All relevant information is automatically retrieved via GIS interfaces via API. As this is a clearly structured decision-making process, a simple if-then logic is used instead of generative AI.

The screenshot shows a web interface for selecting a project type. On the left is a sidebar with four steps: 'Project location' (Enter address), 'Project details' (Select project type), 'Upload documents' (Add documents), and 'Confirmation' (Review & conclusion). The 'Project details' step is active. The main area is titled 'Project details' with the instruction 'Please select the type of your project.' Below this is a section labeled 'Type*' containing eight options, each with an icon and text: 'Solar panels on roofs', 'Solar panels on façades or free-standing' (highlighted with a red border), 'Solar panels in industrial zones (roof, façade, free-standing)', 'Indoor air-to-water heat pumps', 'Outside air-to-water heat pumps', 'Geothermal probe heat pumps', 'Connections to a district heating set', and 'Publicly accessible charging stations'. At the bottom right is a button labeled 'Display ÖREB'.

Figure 7: For the project type «solar panels on facades or free-standing», rule-based checks indicate that a building permit procedure is required.

04. Prototype AI pre-check in the reporting procedure

Step 3: Upload documents and perform preliminary checks

Users upload the documents required by BVV. The prototype automatically checks the entries for completeness. In addition, AI-supported preliminary checks can be carried out to check, for example, the correct scale in the situation plan, the recognisability of a solar installation on the roof surface or the marking of a heat pump. The prototype provides clear feedback on missing or incorrect information.

e

Manufacturer's product description

Product description of the solar system and illustrations of the modules and system components used

✔ 1 file uploaded

Click or drag files here to add more

solar_datenblatt.pdf
1.11 MB

⚙️ Automatic verification

✔ Solar panel installation product description available

The document is a detailed product description and specification for a 'solar module aleo S.19'. It contains performance data, electrical and mechanical specifications, dimensions and quality features of the module.

✖ Illustrations of the modules and plant components available

The document contains photos and illustrations of the solar module itself (page 1: Fots, page 2: technical drawing). However, there are no illustrations or photos of other solar system parts (such as inverters, mounting systems, etc.) available.

✔ Low reflectivity of modules

The document mentions 'High Efficiency Module mit AR-Glas' in the yellow box on the first page. 'AR-Glass' stands for 'anti-reflective glass', which implies low reflectivity

f

Orientation plan according to fire protection data sheet

Orientation plan according to the fire protection [information sheet 'Solar panels' published by the Association of Cantonal Fire Insurance](#)

more

Figure 8: Example of an AI-based preliminary check of a solar system manufacturer's product description with AI-generated feedback on missing modules

04. Prototype AI pre-check in the reporting procedure

Step 4: Check results and submit application

At the end of the pre-check, users can download the results in the form of a report and receive a summary. On this basis, they can complete the application and then submit it to the building permit authority in a quality-assured manner (the interface to the official submission platforms has not yet been implemented).

Conclusion on the prototype

The developed prototype⁸ shows that a targeted combination of rule-based logics and AI-supported preliminary checks in the reporting procedure can create concrete added value. The decisive factor is the differentiated use of technology: Rule-based approaches ensure reliability, while AI helps with the quality check of submissions in particular.

opportunities

- **Rule-based before AI:** Clearly defined decision logics can be mapped efficiently and correctly on a rule-based basis; the use of generative AI is not necessary here.
- **Added value for planners:** Applicants and persons who professionally offer the construction of solar panels, heat pumps or charging stations receive early clarification on the applicable procedure and the basic feasibility of a project at the location.
- **Relieving the burden on municipalities:** If the data is correct, a system can automatically provide the right answers. Rule-based preliminary checks reduce the inspection effort in building authorities, especially in the case of frequently recurring standard cases.
- **Improve the quality of entries:** AI-supported preliminary checks reliably identify many formal errors and incomplete documents; focusing on common deficiencies is particularly effective.

Herausforderungen

- **Probabilistic AI:** Generative AI does not deliver deterministic results; administrative use requires configurations that are as controlled and reproducible as possible, otherwise trust in the AI systems will be lost. Error tolerance is of central importance here: What proportion of correctly executed AI-based pre-checks is necessary to achieve practical benefit? For example, is 95 percent sufficient, or is 99 percent necessary to avoid unnecessary uncertainty in the process?
- **Data availability and legally binding nature:** The degree of automation depends on complete and legally binding data bases; not all municipal inventories (e.g. municipal monument protection) are consolidated in the ÖREB cadastre, and in practice there may always be discrepancies between the ÖREB cadastre and legally binding municipal building and zoning regulations. Information on this is placed in the prototype.
- **Municipal autonomy:** Formal and building law requirements are interpreted differently at local level (e.g. projects within a construction line). A rule-based tool must be able to map these local differences flexibly.
- **Special cases and case-by-case assessments:** Certain projects (e.g. solar panels) are only suitable for a purely automated pre-check to a limited extent. Objects that have not been inventoried

⁸ Interested persons can request test access for the use of the prototype directly from the sandbox team.

04. Prototype AI pre-check in the reporting procedure

may also be worthy of protection and may require a case-by-case assessment (e.g. as part of an ensemble or because of their importance for local development).

- **System integration:** The practical added value of AI only comes from embedding it in existing digitalisation initiatives (e.g. «eBaugesucheZH»). Isolated isolated solutions increase interface costs, media discontinuity and susceptibility to errors, and complicate scaling, operation and acceptance in the building permit process.

05.

Legal framework



What are the basic requirements if AI is to be used in the building permit process?

When using AI in the building permit process, the general requirements of administrative law must be complied with. In particular, the principle of legality must be observed, according to which government action requires a legal basis, whereby the requirements for this legal basis depend on the type and intensity of the use of AI.

The principle of equal treatment also applies, according to which comparable applications must be assessed according to uniform criteria and no objectively unjustified disadvantages may result. The principle of proportionality, the obligation to state reasons and the prohibition of arbitrariness must also be observed. The results supported by AI must therefore be objectively justified, comprehensible and proportionate to the public interests pursued; arbitrary or disproportionate results are not permitted. At the federal level, EMBAG has regulated since 2024 the use of electronic means to fulfil governmental tasks, such as the transfer of electronic administrative assistance activities to private individuals, the disclosure of source codes and the publication of procured and generated administrative data – which cantons and municipalities are increasingly oriented towards.

Does the client need to know that AI is being used?

Whether the client is required to provide information depends on the specific use of AI. Under certain circumstances, such an obligation may arise from data protection law where the processing of personal data is concerned. However, there is no general obligation of transparency regarding the way in which the administration organises its internal work processes.

However, an information and disclosure obligation can be triggered when a decision is made entirely or predominantly by an AI system without sufficient human scrutiny. In this context, the Canton of Zurich is currently working on a transparency register to identify which algorithmic decision-making systems are used that potentially have an impact on the fundamental rights of individuals (Directory of algorithmic decision-making systems [AES]). The legal basis for this is being created as part of the revision of the cantonal law on information and data protection (IDG ZH).

⁹ Federal Act on the Use of Electronic Means for the Completion of Public Authorities (EMBAG) of 17 March 2023 (SR 172.019).

05. Legal framework

Responsibility and decision-making authority

Can AI decide whether a project is subject to notification or approval?

The decision as to whether a project is subject to notification or approval must be made by the competent authority. It must not delegate it to an AI system. The authority to issue directives in the planning permit procedure generally lies with the municipal building authority. AI may only be used to support decision-making, e.g. to structure facts or prepare legal evaluations. Fully automated decision-making by AI, on the other hand, is not permitted without a legal basis.

Administrative decisions must comply with constitutional and administrative procedural requirements. This includes the obligation to give reasons for a decision. This is based on the right to be heard pursuant to Art. 29 BV. If a decision is prepared using an AI system or influenced by it, the underlying logics must be comprehensible and explainable in their essential features. AI systems must therefore not deliver non-transparent black box results whose content cannot be verified by the parties or by appeal bodies.

«AI can support building authorities, but the justification and responsibility for decisions remain with humans»

Stephanie Volz, Managing Director ITSL, University of Zurich

Where can AI be used meaningfully from a legal perspective in planning permission or registration procedures and where not?

Areas where AI can be used with few legal problems include checking the completeness of submitted documents and automated comparison with relevant standards. AI can also be used sensibly to identify risks, structure issues and prepare official decisions.

However, the use of AI for autonomous decision-making is not permitted, particularly when a project qualifies as subject to notification or approval or when a request is materially rejected. Such decisions remain the responsibility of natural persons and may not be delegated to technical systems under current law.

Does it make a difference from a legal perspective whether rules-based systems or generative AI systems are used?

No, in principle, the same legal requirements apply to rule-based systems and generative AI systems. In practice, however, the use of rule-based systems is likely to be much simpler, as legal requirements such as the obligation to provide reasons can be complied with much earlier due to the easier traceability. The use of generative AI can also raise further questions, particularly with regard to risks related to explainability, bias, third-party worker participation and governance.

Data protection

What requirements must be observed from a data protection point of view?

A great deal of personal data is processed during the planning permit and registration process, including the names of the parties involved, ownership structure and any objections raised by third parties.

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The provisions of cantonal data protection laws apply to this data processing. The IDG ZH applies in the canton of Zurich.

First of all, the principle of purpose limitation is key: Personal data may only be processed for purposes arising from the planning permission process. Further use, e.g. for training AI systems, is only permitted under special circumstances, e.g. if there is a corresponding legal basis. In addition, the principle of data minimisation must be observed, according to which only the data required to carry out the reporting procedure may be processed.

Is there an entitlement to human verification through AI-supported preliminary testing as part of the building permit or registration process?

A preliminary review, such as checking whether a notification or application is complete, is prior to the actual decision. The Swiss Federal Act on Data Protection (FADP) provides special protection mechanisms for fully automated individual decisions with legally significant effect. Under Art. 21 FADP, data subjects have the right to be informed of such a decision and to request a review by a natural person. The prerequisite is that the decision is purely automated and has significant legal or comparable consequences for the data subject. Whether decisions made as part of the planning permit or registration procedure reach this materiality threshold will need to be examined on a case-by-case basis. The decision as to whether the documents submitted are complete is unlikely to reach the threshold, and the decision as to which procedure should be carried out is unlikely to reach the threshold. The same applies to the granting of a building permit or the rejection of a building application. Such fully automated decisions would likely not be permissible under current law.

There are no corresponding data protection regulations at cantonal level, particularly in the canton of Zurich. However, such a right could arise indirectly from several overriding legal bases, namely the constitutional right to a fair hearing, the principle of the rule of law and the general provisions of data protection law.

In the planning permission or registration procedure, it must also be taken into account that some decisions in the course of the procedure may be qualified as dispositions. If an order is in fact issued by an AI system, it must be ensured that the responsible authority can review its content and at least justify it upon request.

In summary, although there is no explicit special provision in cantonal law, there is an obligation derived from federal and constitutional law for human review as soon as AI systems independently make decisions with legally relevant implications in the planning permit or registration process, which, under current law, would likely not be permissible anyway. Purely automated, non-verifiable process control would not be compatible with these principles.

Procurement

What legal issues arise when authorities procure AI systems downstream in the building permit process?

Consideration of public procurement of AI systems by public authorities also raises legal questions that need to be clarified on a case-by-case basis. Important questions to be clarified relate to suitable award criteria such as the explainability, bias mitigation and auditability of AI systems, particularly in the case of black-box models. Such criteria must be measurable, non-discriminatory and proportionate. Similar to

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the awarding of public clouds,¹⁰ requirements on the location of data centres in Switzerland and geo-redundant services can also play an important role in the procurement of AI models.

Responsibility and liability

Who is liable for errors in building permit or registration procedures caused by AI?

If an incorrect AI-supported recommendation leads to an unlawful decision in the planning permit or registration procedure, the relevant public body, i.e. the state or municipality, could in principle be held liable based on the relevant state liability standards. Responsibility for the error lies solely with the deciding authority; as a rule, direct liability of the software provider towards the affected parties does not exist or is legally excluded.

The use of an AI system does not relieve the authority of its responsibility. Even if the error was technically caused, the decision should be legally attributed to the Authority. Internally, however, the public sector can check whether contractual or non-contractual recourse claims exist against the software provider, for example due to breach of contractual obligations.

In general, please note: Today, the administration enjoys a great deal of trust amongst the population. However, this trust could be undermined by the use of non-transparent AI systems or by wrong decisions. Transparent communication, a clear presentation of the role of humans and machines, and – where possible – an emphasis on voluntary action and on opportunities for control and legal redress are key to avoiding a loss of acceptance and legitimacy.

What if the government offers voluntary AI-supported pre-checks to check, for example, whether the submitted documents are complete?

A state responsibility is not excluded even in the case of voluntary offers. However, the issue needs to be viewed in a differentiated way: Pre-checking is not an order. However, this may give rise to a trustworthy liability. If the pre-check gives the impression that it is reliable, binding or authoritative, users can gain a level of trust worthy of protection. It is therefore important to clarify that the pre-check is non-binding and that the relevant verification takes place in the formal procedure by the Authority.

What happens if an AI system incorrectly classifies a project requiring approval as permissible?

If an AI system incorrectly classifies a project as permissible as part of the building permit or registration process and the authorities do not decide otherwise, this constitutes an error on the part of the authorities, even if the cause is technical. The resulting legal situation is attributable to the competent authority.

If the client acted in good faith and was allowed to rely on the legality of the procedure due to official conduct – including the handling of proceedings influenced by AI – it may, under certain circumstances, invoke the constitutional protection of legitimate expectations. This can be particularly effective if the Client has already taken measures based on the decision that cannot be reversed without disadvantage.

At the same time, it should be noted that the protection of legitimate expectations has its limits where the illegality was obvious or where the building owner itself should have realised that the requirements of the notification procedure were not met.

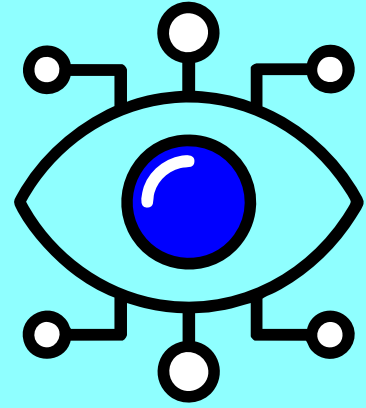
¹⁰ «Public clouds» allocation: Contracts can be concluded, press release on judgement B-3238/2021, dated 20 October 2021.

05. Legal framework

In summary, the legal chapter shows that the use of AI in building permit and registration procedures is legally possible as long as it is supportive, transparent and under the full responsibility of the authorities – and thus sets the framework for the legal and factual conclusions. It should be stressed, however, that the present legal explanations serve only as a first orientation on the possible legal issues. Before the actual use of an AI application, an in-depth legal clarification must always be carried out.

06.

Conclusion and outlook



The report summarises the key findings of the AI sandbox project and highlights the conditions under which AI can meaningfully support the building permit process and the next steps that are required at a technical, organisational and political level.

1. Practical findings

The project systematically analysed the potential and limitations of AI in the building permit process and tested them in practice using a functional prototype for an AI-supported preliminary check in the registration process.

2. Clear prerequisites for deployment

Productive use of AI requires reliable and legally binding data bases, structured submissions, modular system architectures, comprehensive test and evaluation pipelines and clear governance structures.

3. Rule-based versus generative

AI is not an end in itself. Where decision logics are clearly defined, rule-based checks should be used. Generative AI is primarily suited to supporting and quality assurance tasks such as evaluating text documents.

4. No full automation

Decisions and responsibilities remain with people; complete automation of the planning permission process is currently unrealistic. Instead, individual test steps can be specifically supported, for example through preliminary checks to improve the quality and completeness of building applications.

5. Connectivity of the solution

The sandbox prototype is designed in such a way that it could be further developed by external actors and converted into productive software, in particular by integrating it into existing digitisation initiatives (e.g. «eBaugesucheZH» or GIS-Browser) as well as through cross-community solutions.

6. Profitability and technological perspective

The economic benefit of AI lies in supporting recurring testing steps and improving quality throughout the process. For a realistic valuation, implementation, integration and control costs must be taken into account in addition to the pure usage costs. However, with increasing technological advances and better embedding in existing systems, there is good potential to make the use of AI more economical and broadly usable in the future.

7. Contribution to the political debate

The results of the AI analysis and the development of the prototype will make a concrete, evidence-based contribution to answering political initiatives such as AI in the building permit process in the canton of Zurich. Similar initiatives in other cantons underscore the national relevance.

8. Simplification of the regulatory framework

AI cannot eliminate the complexity and fragmentation of building regulations at the cantonal and municipal level and limits the use of appropriate systems. Existing trade-offs require political clarification and cannot be resolved technologically.

06. Conclusion and outlook

9. Consider the overall context

The building permit procedure is part of a longer planning and implementation process. Efficiency and quality gains can only be achieved through a holistic approach, such as digital assistants throughout the entire process, planning tools with integrated review of legal requirements, or BIM-based models with structured data. Measures to overcome the fragmented system and application landscape help to implement AI systems and automation in a targeted manner without media discontinuity.

10. Open source and open interfaces

For the sustainable development and wider use of AI-based solutions, the canton could make selected technical building blocks – such as test rules, reference implementations or interfaces – available as [open-source repos](#) and accessible via standardised APIs. Such an approach promotes transparency, reusability and cooperation with communities, software providers and the planning industry, without imposing production systems or shifting central control.

The sandbox results show that AI can support the building permit process in a targeted and effective way, as long as its use is clearly limited, technically embedded and organisationally secured. The greatest added value today does not lie in the automation of decisions, but in improving the quality and completeness of requests, in reducing the burden of recurring review steps and in better access to information. At the same time, the report makes it clear that legal fragmentation, discretionary decisions and conflicting policy objectives cannot be overcome by technological solutions. Long-term benefits can only be achieved through legal development, consistent digitalisation throughout the entire construction process and the responsible use of AI.

Glossary

API (application programming interface)

Interface through which software systems can communicate with each other in a standardised way and exchange functions or data without disclosing the internal logic.

Building Information Modelling (BIM)

Digital, model-based method for planning, executing and managing buildings based on a common, structured 3D building model.

Chatbot

AI-supported dialogue system that processes user input in natural language and automatically generates responses or suggestions for action.

GIS browser

Digital map application that visually displays geographic information and spatial data (e.g. zones, protected areas or pipelines) and often serves as one of the bases for clarification to determine the correct procedure (e.g. reporting procedure).

Large Language Models (LLMs)

AI models that have been trained with large amounts of text and can analyse, understand and generate natural language, for example for summaries, queries or draft texts.

LLM-as-a-Judge

Approach in which a large language model is used to evaluate AI outputs, e.g. to assess quality, consistency or compliance with defined criteria.

Reporting procedure

Simplified building law procedure in which a project is notified to the authorities and may be implemented without a formal building permit, as long as the legal requirements are met.

Noise

Distracting or irrelevant information in data or model outputs that may affect the accuracy, comprehensibility or reliability of the results.

Open-source repositories

Publicly accessible repositories for source code or technical components under open licences that allow third parties to view, use and further develop without specifying a binding use.

ÖREB (Register of Restrictions on Ownership under Public Law)

Publicly available information on legal restrictions on properties, e.g. due to building, environmental or monument protection regulations.

Retrieval-augmented generation (RAG)

Approach in which a language model calls up external, verified sources of information and incorporates them into text generation in order to increase timeliness and accuracy of facts.

Semantic search method

A search method that finds content not only by keywords, but also by their meaning and context.

Zero-shot approach

Ability of an AI model to work on tasks or answer questions without having previously been specially trained for this use case.

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Case studies from the Innovation Sandbox for AI

The «AI in building permits» project served as a case study within the innovation sandbox for AI in the canton of Zurich. **NOKEMA** submitted the project proposal and implemented it together with technology partner **Byte Studio**, specialist partner **Gossweiler Ingenieure AG** and implementation partner **Stadt Kloten** as part of the Innovation Sandbox. The project team analysed specific AI use cases throughout the building permit process and developed a prototype for an AI-supported preliminary check in the reporting process. The findings and recommendations of this report are based on the experience of this case study.

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